Individual Assignment 5

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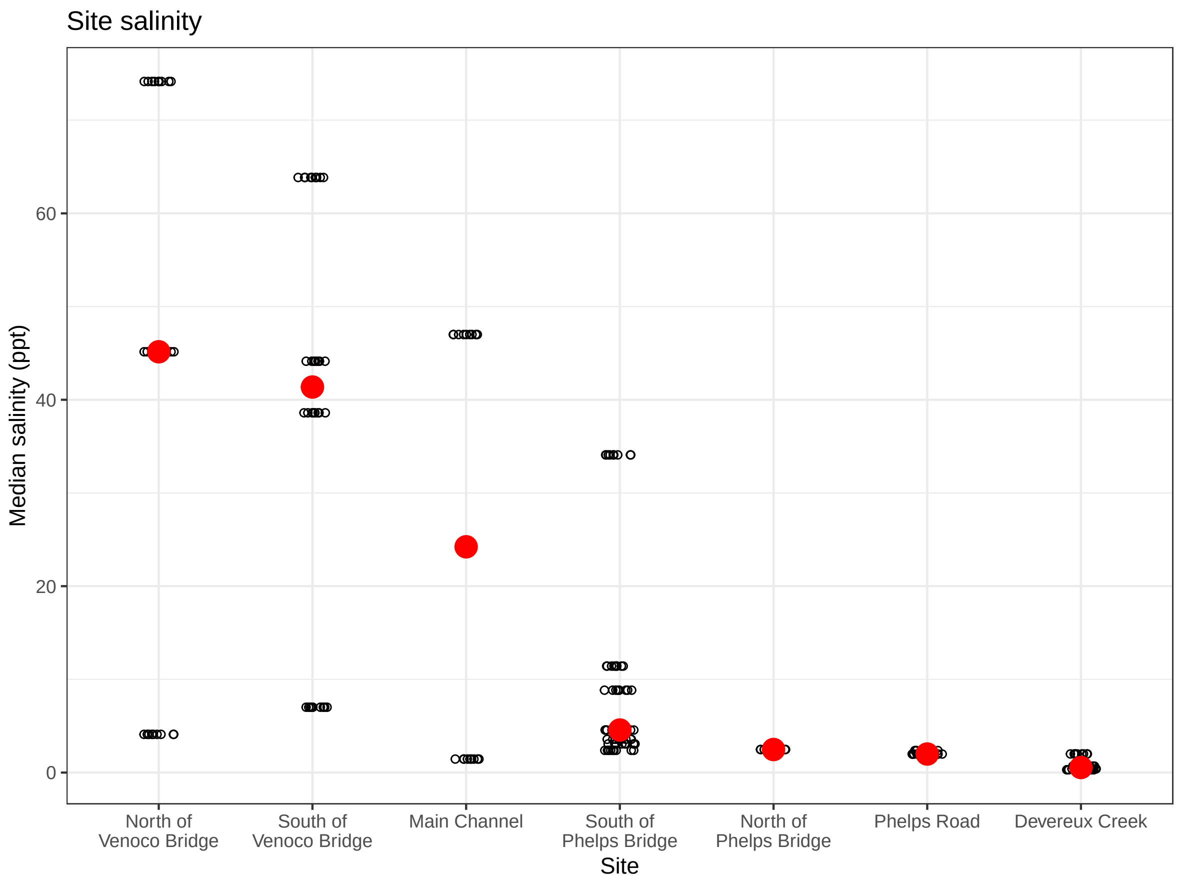
1. Set up

Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

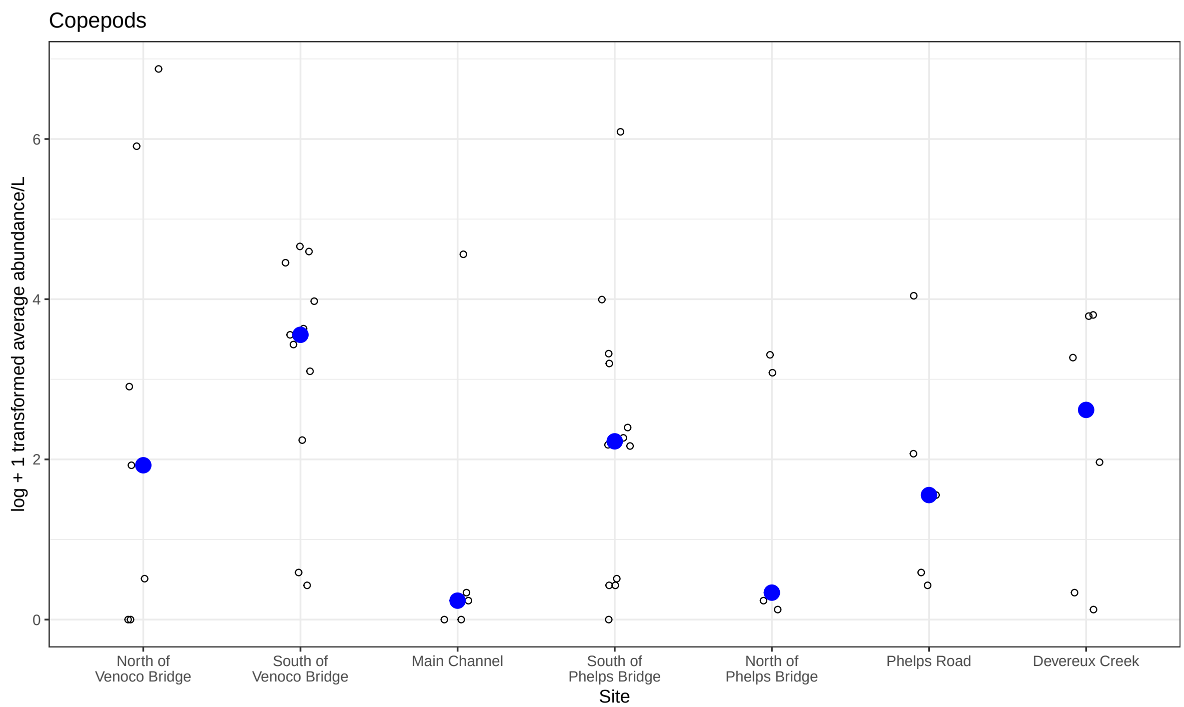
Data

Importing other graphs




```
# base layer
copepods_plot <- ggplot(data = copepods_df,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
```

```
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +
  # different theme
  theme_bw()

# displaying object
copepods_plot
```



Question 1

A: Planning

- x axis: Site

- y axis: log transformed Ostracod abundance
- geometry abundance: geom_jitter()
- geometry median: stat_summary()

B: Wrangle Data

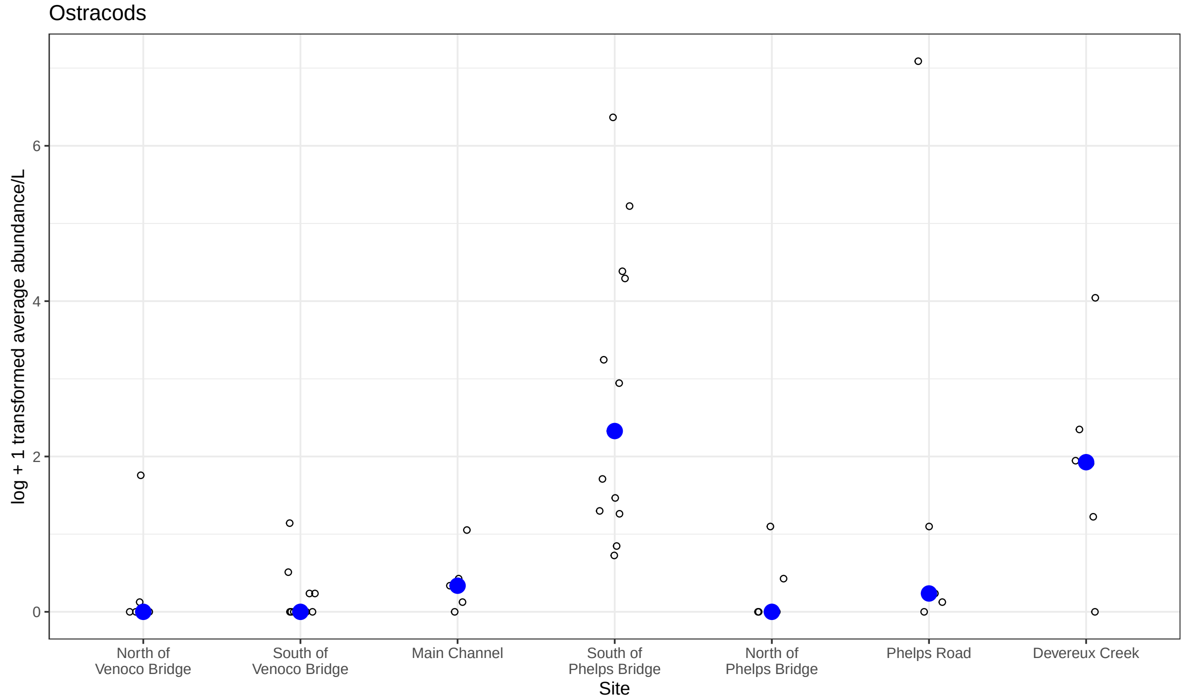
```
#new object from aq_ins_clean
ostracod <- aq_ins_clean |>
  # filter to only include ostracod
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows
slice_sample(ostracod, n=10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2022-05-05_NEC  2022-05-05 00:00:00 NEC   ostr~   0.267   9.45  63.8     7.3
 2 2023-03-03_NPB2 2023-03-03 00:00:00 NPB2  ostr~   0.133   7.71   1.98    9.59
 3 2023-08-12_NEC  2023-08-12 00:00:00 NEC   ostr~   0      NA     NA      NA
 4 2023-05-04_NEC  2023-05-04 00:00:00 NEC   ostr~   0.667   8.05  NA       8.55
 5 2023-02-12_NEC  2023-02-12 00:00:00 NEC   ostr~   2.13   NA     NA      NA
 6 2022-02-23_NPB1 2022-02-23 00:00:00 NPB1  ostr~   0      NA     NA      NA
 7 2023-02-28_NPB1 2023-02-28 00:00:00 NPB1  ostr~   0      NA     NA      NA
 8 2023-05-05_NPB  2023-05-05 00:00:00 NPB   ostr~  24.7    NA     NA      12.1
 9 2022-05-11_NDC  2022-05-11 00:00:00 NDC   ostr~   0       9.11   0.398   9.4
10 2023-11-12_NMC  2023-11-12 00:00:00 NMC   ostr~   0.533  NA     47      14.0
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

C: Code Figure

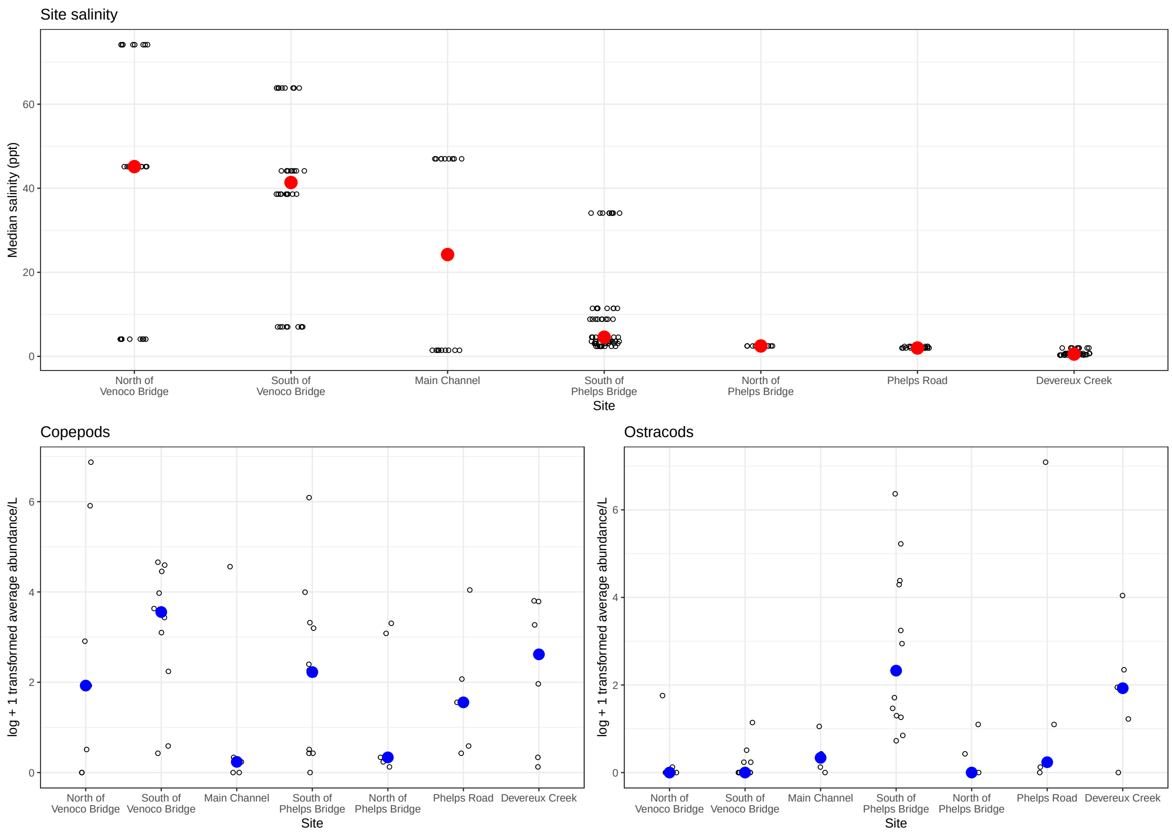
```
# base layer
ostracods_plot <- ggplot(data = ostracod,
```

```
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # different theme
  theme_bw()

# displaying object
ostracods_plot
```



D: Create multipanel figure

```
#combining plots
salinity/(copepods_plot+ostracods_plot)
```



E: Analysis

- The differences in sites between copepods and ostracod abundance are mainly at the South of Phelps Bridge as ostracods are the most abundant here while copepods are more abundant South of Venoco Bridge.
- Ostracod abundance may not be related to site-level differences as there are numerous occassions of median salinity being relatively low at places like Phelps Bridge or Phelps road, yet South of Phelps Bridge sites have high abundance while the North side doesn't. Additionally,low abundance sites like North of Phelps have the same abundance as high salinity sites like Venoco Bridge.

Question 2

A: Planning

- x axis: salinity

- y axis: log transformed abundance
- color: taxon
- geometry: geom_point()
- facets: facet_Wrap(~taxon, scales="free")

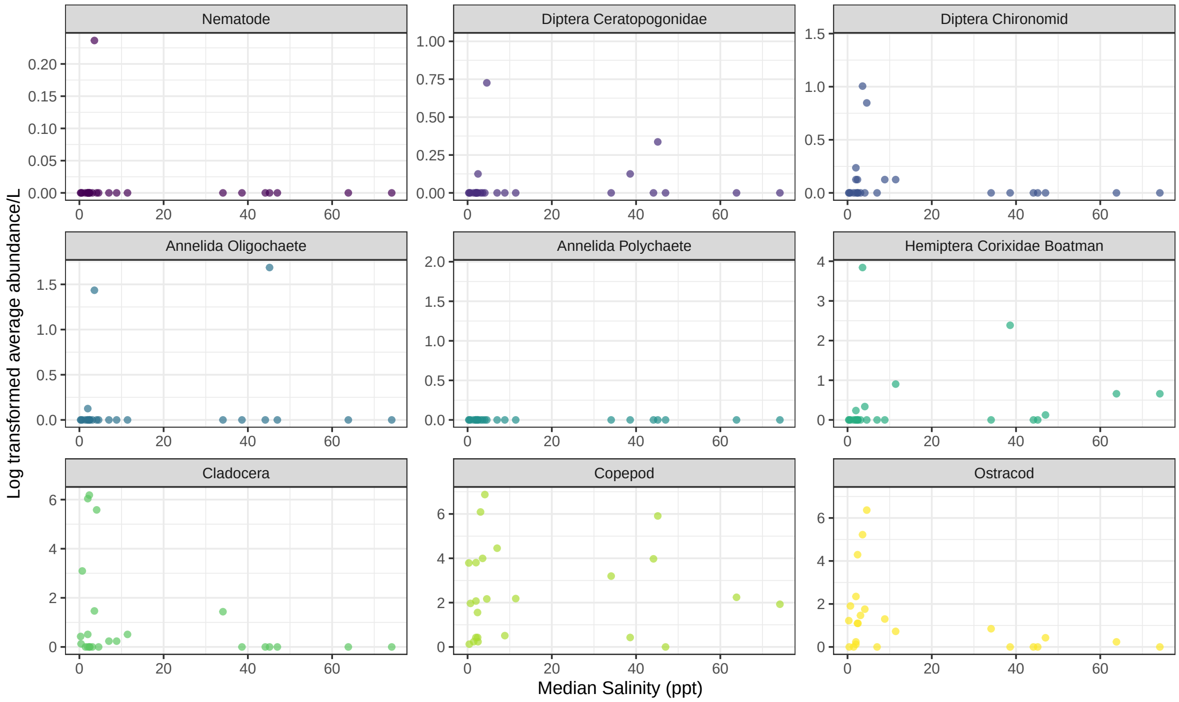
B: Wrangle Data

```
# creating new clean object from aquatic inverts
ab_salinity <- aq_ins_clean |>
  # log transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit+1),
         # convert taxon to title
         taxon = to_any_case(taxon, case = "title"),
         # order taxon levels by abudance
         taxon = fct_reorder(taxon, ave_lit, .fun="max"))

#display 10 random rows
slice_sample(ab_salinity, n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <fct>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2022-02-23_NPB1 2022-02-23 00:00:00 NPB1  Nema~   0      NA     NA      NA
 2 2023-07-17_NPB  2023-07-17 00:00:00 NPB   Anne~   0      NA     NA      NA
 3 2022-01-28_NDC  2022-01-28 00:00:00 NDC   Anne~   0       7.65   2.00    9.55
 4 2021-08-25_NPB  2021-08-25 00:00:00 NPB   Clad~   0       7.52   4.57    4.5
 5 2023-11-12_NMC  2023-11-12 00:00:00 NMC   Clad~   0      NA     47      14.0
 6 2022-05-11_NDC  2022-05-11 00:00:00 NDC   Dipt~   0       9.11   0.398   9.4
 7 2023-03-03_NDC  2023-03-03 00:00:00 NDC   Cope~   6.13    7.45   0.688   5.64
 8 2022-02-21_NPB  2022-02-21 00:00:00 NPB   Dipt~   0       7.39  34.1     0.34
 9 2023-01-28_NPB1 2023-01-28 00:00:00 NPB1  Anne~   0.133   7.66  NA       9.74
10 2023-09-15_NMC  2023-09-15 00:00:00 NMC   Ostr~   0.4    NA     NA      14.2
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

C: Code figure

```r
# base layer
ggplot(data = ab_salinity,
       # x-axis: site
       # # y-axis: mean abundance/L
                   mapping = aes(x = med_sal,
                                 y = log_ave_lit,
                                 color = taxon)) +
  # layer 1: points to represent observations
  geom_point(alpha = 0.7) +
  #facet wrap for taxon
  facet_wrap(~ taxon,
             scales = "free") +
  # use color palette
  scale_color_viridis_d() +
  # labels
  labs( x = "Median Salinity (ppt)",
        y = "Log transformed average abundance/L") +
  # theme
  theme_bw()+
  # remove legend from theme
  theme(legend.position = "none")
```



D: Analysis

- The differences between taxa in relationships between abundance and salinity is that the higher the salinity, the lower the average abundance. However, this isn't the case with all species as Copepods are abundant at high and low salinities, similar to Hemiptera as both show up to 2.5 log average abundance/L in higher salinities. On the other hand, species like Annelida Oligochaete, along with the majority of the species, follow an exponential decrease as salinity increases. This includes Cladocera and Diptera Chrionomid to name a few.